

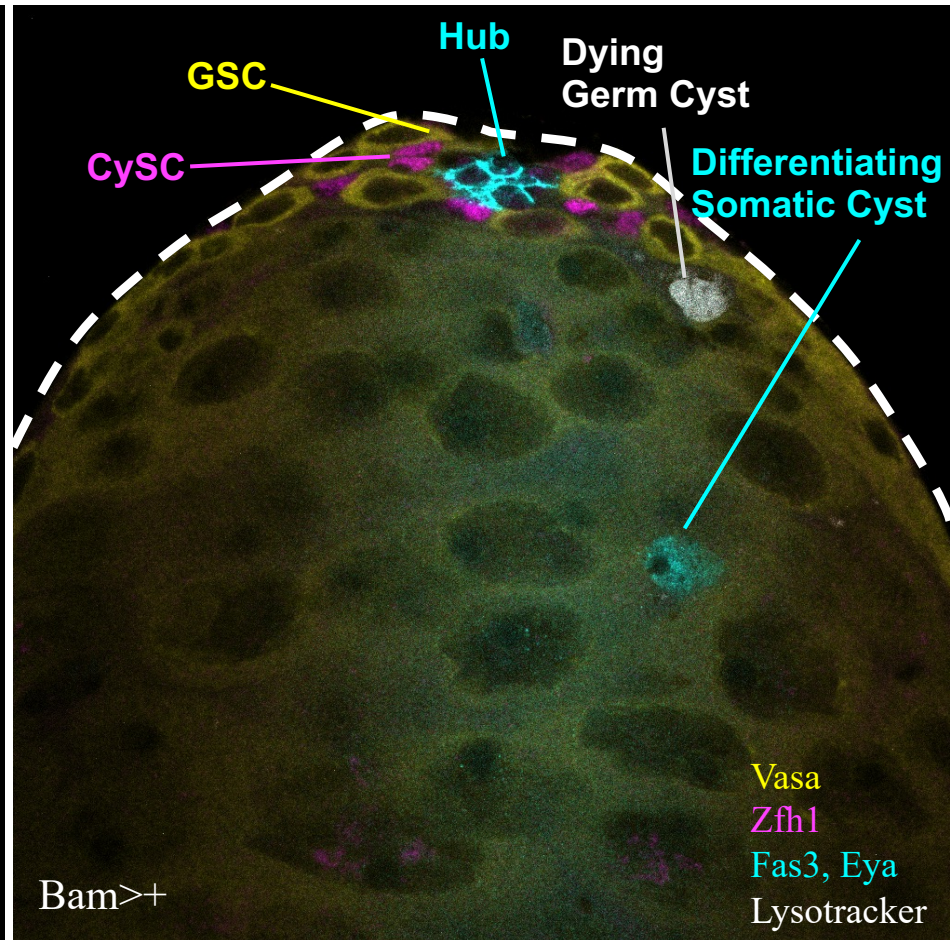
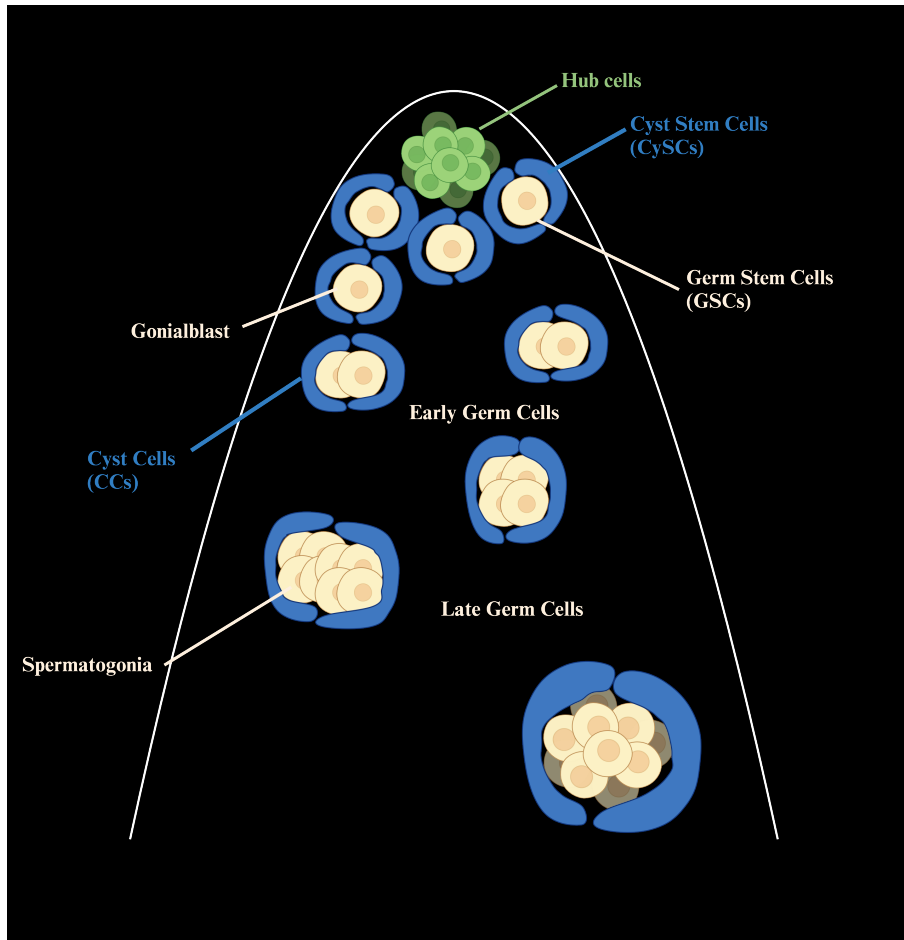
The Role of Monocarboxylate Transporters in Germ Cell Survival in *Drosophila* Testis



Zheheng Fan

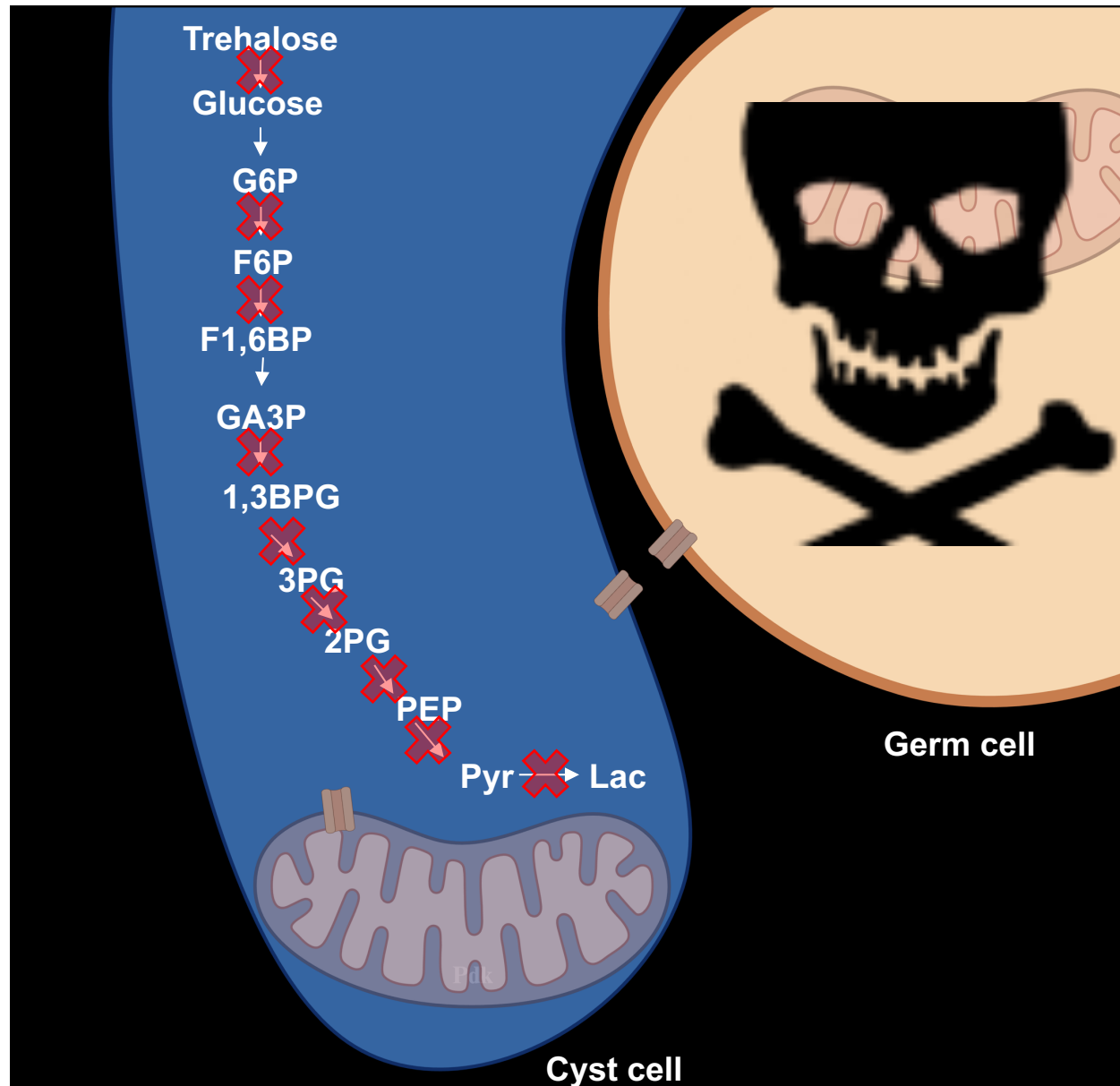
Biomedical Sciences *BSc.*

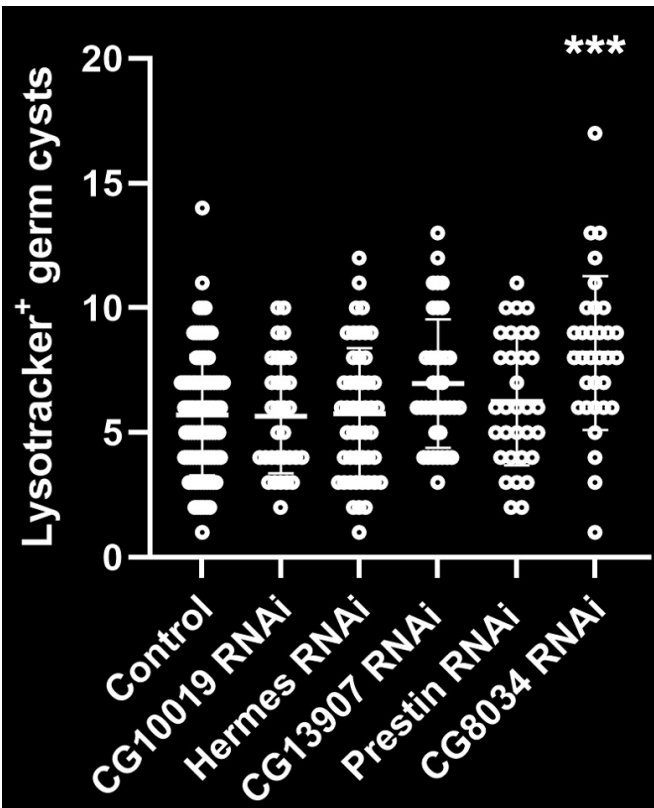
Supervisor: Dr. Marc Amoyel



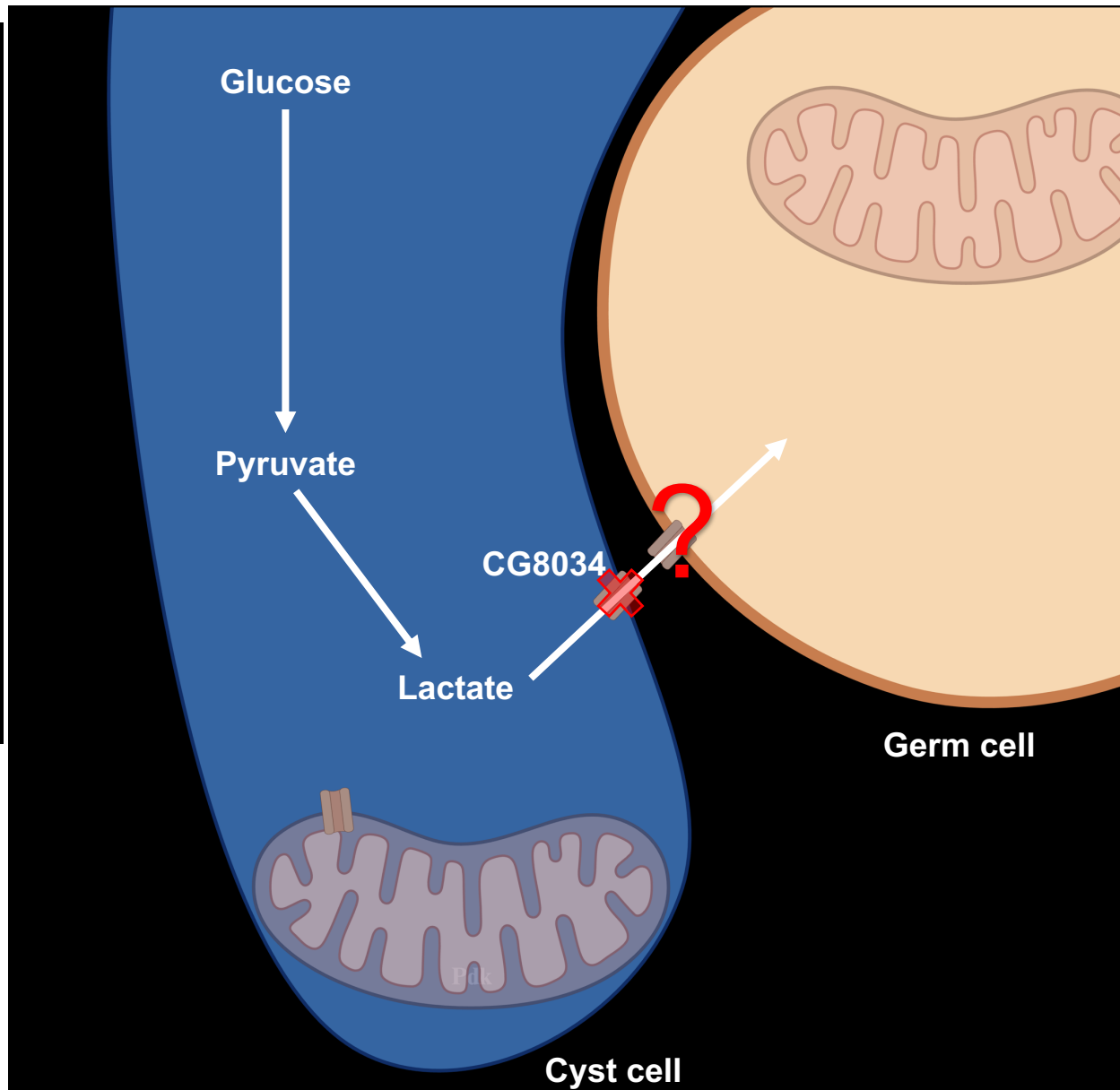
- Cell-cell communications between different lineages coordinate differentiation and maintain tissue homeostasis
- The cell-cell communication can also be metabolic

- Knockdown of glycolytic enzymes in the somatic cells increases germline deaths
- Knockdown of lactose dehydrogenase increases germline deaths

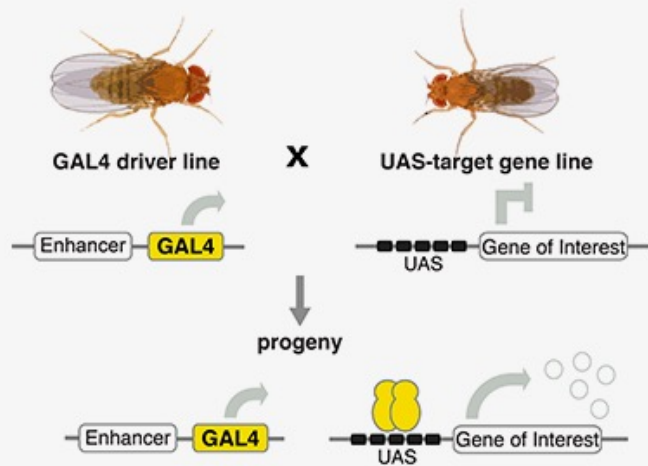




- CG8034 knockdown significantly increases germline deaths

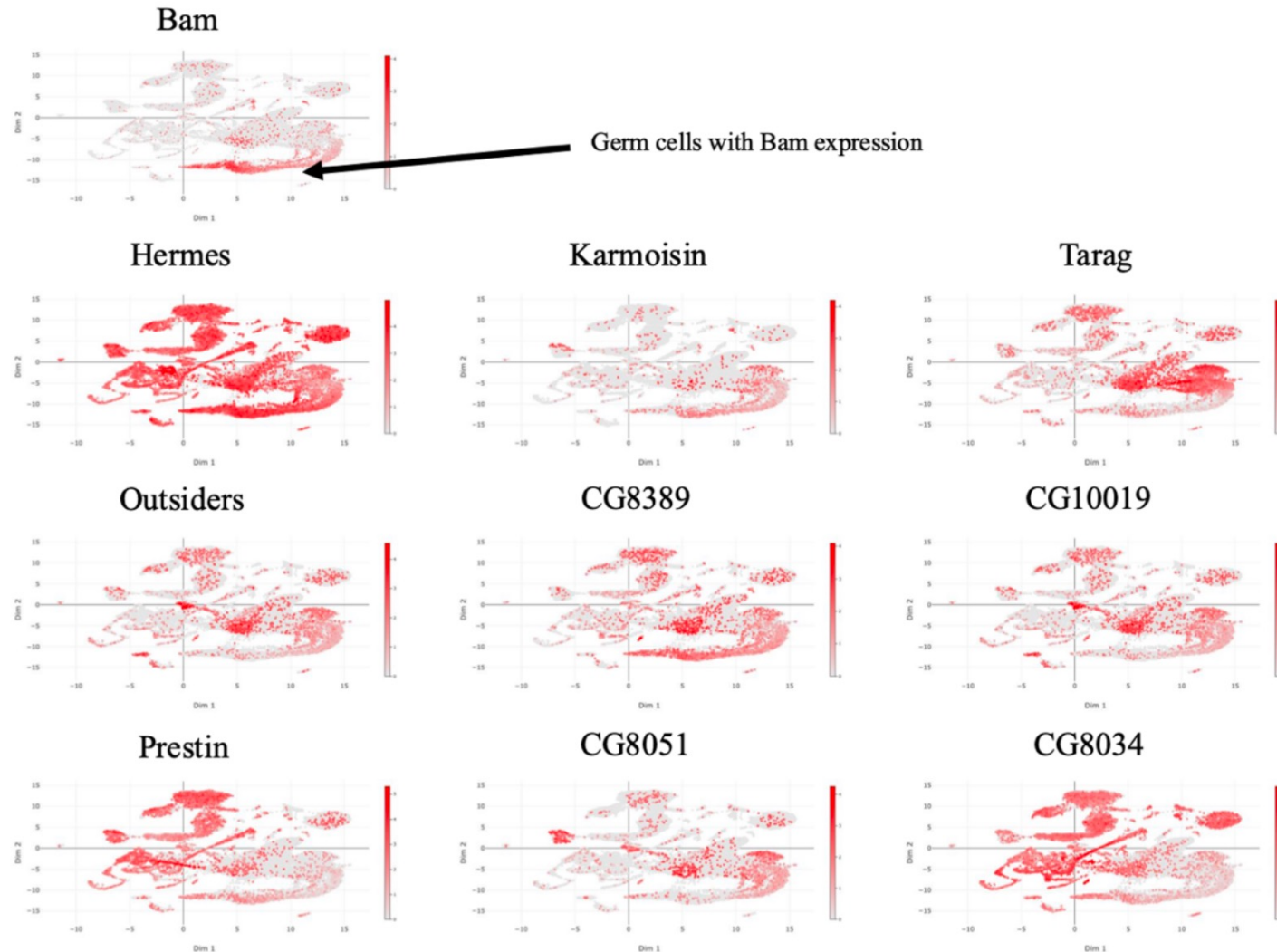


GAL4 System



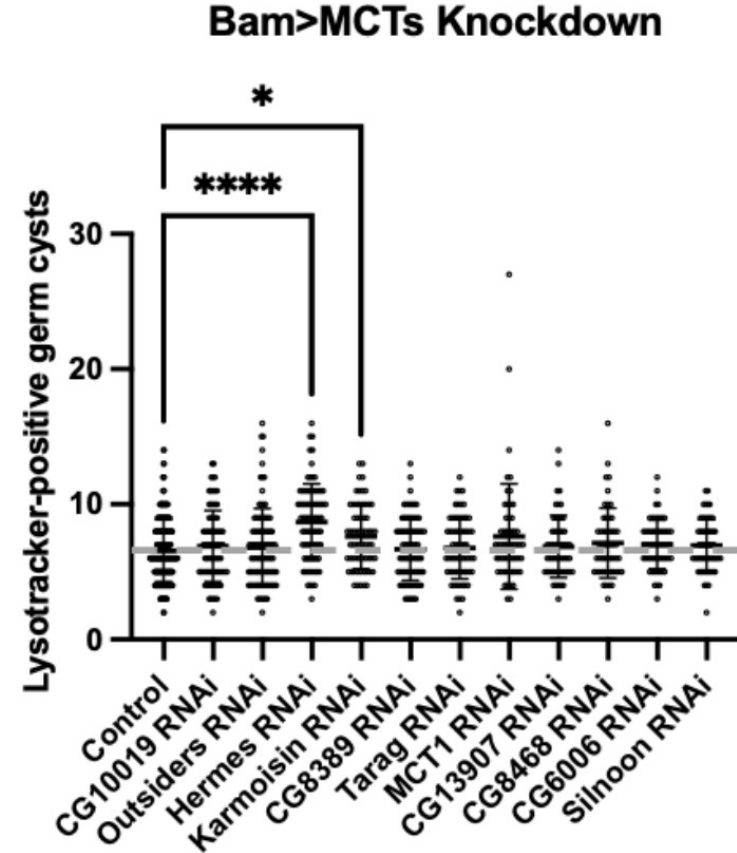
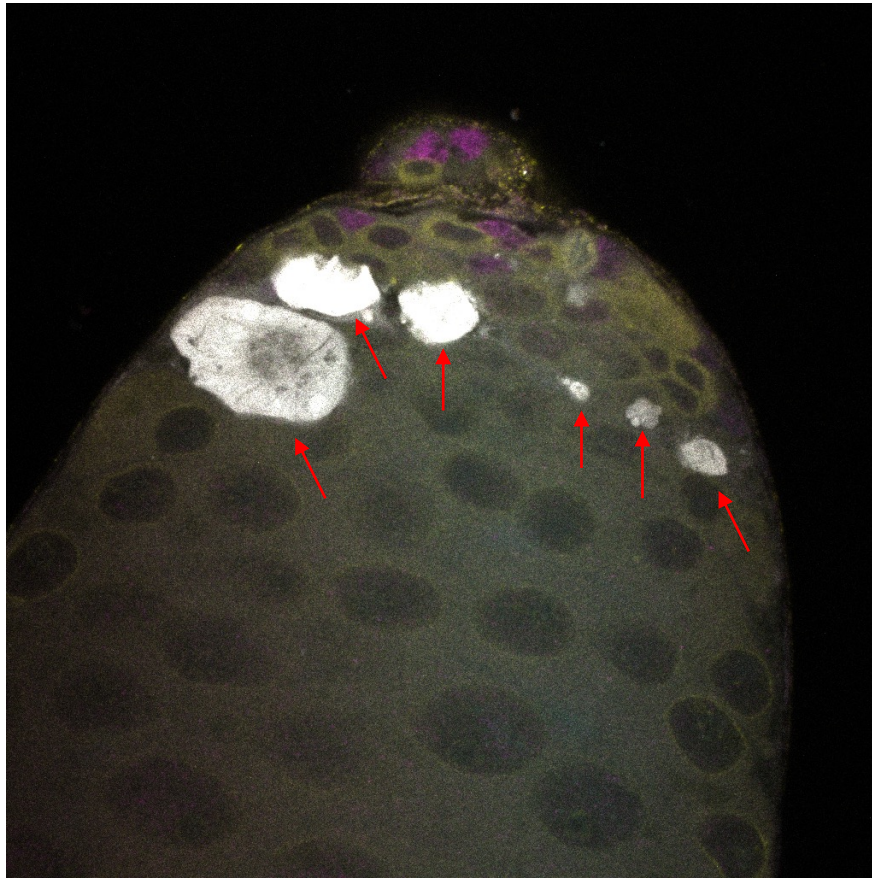
- Gal4 – a transcription factor that binds to UAS to activate gene expression
- Gal4 promoter – drives Gal4 expression in specific tissue
 - Examples: Bam-Gal4, Tj-Gal4, etc

9 MCTs are expressed in the germline



Data from Fly Cell Atlas

- UMAP plot scales down multidimensional single-cell RNA-seq data to 2 dimensions
- Bam (Bag of marbles) is an early germline marker



- Karmoisin and Hermes are primarily identified as an aromatic amino acid transporter and a proton-coupled pyruvate transporter, respectively
- Further validation using the lactate reporter and a second RNAi line is required

Thank you!

Q&A

